Group/LAN

Netweave/Frankfurt-224201565
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REQUEST FOR CHANGE DOCUMENT

Change Description:

1. Change the structure to be 3 Tier Architecture 2.
Add 2 Cisco 4331 Integrated Service Routers.
3.Add 2 Cisco 3650 Multilayer Switches
4.Add 2 Cisco 8500 Series Wireless Lan Controllers
5.Add 1 Cisco UCS C240 M6 Rack server.
6.Add 3 Access/Layer-2 switches (48 ports) 7. Add 120 Cisco Aironet 3800 Series (LWAP) 8. Add 80 Cisco 7960 IP phone.
9. Use Zone-Based Firewalls (configured on routers)
10. Add PCs

Network Addressing Schem

We have chosen to implement the Class B private IP address range 172.0.0.0/18 for our network infrastructure. To efficiently accommodate the required number of hosts, we have subdivided this address space using a subnet mask of /24. This allows for the creation of multiple subnetworks, each capable of supporting up to 254 usable host addresses. By employing this subnetting strategy, we ensure optimal address allocation within our medium to large-scale network deployment, adhering to established networking standards and best practices.

- 172.18.10.0/24 Vlan 10 Accounting & Finance
- 172.18.20.0/24 Vlan 20 ICT Services
- 172.18.30.0/24 Vlan 30 Human Resources (HR)
- 172.18.40.0/24 Vlan 40 Executive Management
- 172.18.50.0/24 Vlan 50 Legal
- 172.18.60.0/24 Vlan 60 Sales & Marketing
- 172.18.70.0/24 Vlan 70 Operations
- 172.18.80.0/24 Vlan 80 Research & Development
- 172.18.90.0/24 Vlan 90 Guest
- 172.18.99.0/24 Vlan 99 Native & Management
- 172.18.110.0/24 Vlan 110 Voice
- 172.18.115.0/24 Vlan 115 BerlinWifi

Change Reason:

1. Change the structure to be 3 Tier Architecture

The initial structure has no redundancy which can cause a single point of failure and negatively affect the network performance, making the network unreliable.
It also presents scalability challenges due tightly coupled intermediary devices which can limit flexibility. Its lack of segmentation also increases the risk of unauthorized access and data breaches, this can lead to complex troubleshooting.

By using the 3-Tier architecture we will be able to address the issues presented by the initial structure, we will segment the network into 3 layers (Core, distribution and Access) , this will improve security, manageability, scalability. simplify troubleshooting for better performance making the network reliable.

2. Add Cisco 4331 Integrated Service Routers.

On the initial structure we have two Cisco 2811 Integrated Services Router which is now obsolete. It also limits scalability.
The 2811 router may require more frequent maintenance and updates, increasing operational expenses over time as it is an older device.

transitioning to the 4331 router aligns with the principles of the 3-Tier architecture by providing more capable and modern networking infrastructure. It enhances performance, scalability, and flexibility, enabling the network to meet current and future demands more effectively. It also ensures better integration with other network components and technologies, supporting the overall goals of the network design.

3. Add 2 Cisco 3650 Multilayer Switches

According to 3-Tier architecture structure, on the distribution layer there must be a device(s) that will be able to transport data from access to core layer and vice versa, we chose to use multilayer switches because of its dual nature.
This will enhance scalability, redundancy and improve performance which will result into simplified network management.

4. Add a Cisco 8500 Series Wireless Lan Controller

Adding a wireless Lan controller will provide a centralized management and control of access points throughout the network. This will simplify the configuration, monitoring and troubleshooting, enhancing overall network management efficiency.

A WLC offers advanced security features such as encryption, authentication, and intrusion detection/prevention systems. Centralized security policies can be enforced across all wireless devices, ensuring consistent protection against potential threats.

5. Add 1 Cisco UCS C240 M6 Rack server (backup server.)

By adding a backup server to the network design, it adheres to the 3-tier architecture by enhancing data redundancy, fault tolerance, disaster recovery capabilities, and overall data protection. This addition supports the network's objectives by ensuring the availability, integrity, and reliability of critical data and services.

6. Add 1 Access/Layer-2 switches (48 ports)

The initial structure has 2960 switches with 24 ports which limits scalability.

To cater for future growth of 40% we are going to deploy layer 1 switch with 48 ports on the access layer. This conforms to the 3-tier architecture by enhancing segmentation, scalability, performance, and manageability.

7. Add 6 Cisco Aironet 3800 Series (LWAP)

Adding access points per department on the Access layer will improve wireless coverage. This ensures that users within each department have reliable access to the network without experiencing dead zones or areas of poor signal strength. This addition enhances segmentation, scalability, performance, and security, supporting the network's objectives and ensuring efficient operation across departments.

8. Add 80 Cisco 7960 IP phone.

Adding IP phones per department in a 3-tier network design improves communication quality, scalability, and security. By organizing voice traffic by department, the network ensures clear calls and efficient performance. It also makes it easier to expand the system as the organization grows and keeps voice conversations safe from unauthorized access.

9. Add zone-based firewalls (Configured on Cisco 4331 routers)

We decided to opt for zone-based firewalls instead of dedicated firewalls to streamline security across all our regions. With zone-based firewalls, security policies are applied based on traffic flow between defined security zones rather than relying on separate dedicated hardware firewalls. This approach enhances scalability, flexibility, and ensures consistent security across all locations (Berlin, Munich, etc.) in the company. It also reduces complexity and costs associated with maintaining multiple firewalls, while centralizing control and improving monitoring of security policies. Zone-based firewalls provide the ability to implement detailed, granular policies and prioritize different types of traffic within the same network, which optimizes performance and strengthens security.

Impact of Change:

The change involves a big revamp of the network structure to fit the 3-tier architecture, fixing issues like lack of backup, outdated devices, and limited scalability. We're adding modern routers, switches, a wireless LAN controller, a backup server, and more access points and switches per department. These updates make the network more reliable, scalable, and secure, ensuring smooth operation for both current and future needs.

Routers (Cisco 4331 Integrated Service Routers): These provide more modern and efficient routing capabilities compared to the old setup. They enhance scalability and performance, ensuring that the network can handle increasing traffic demands and better integrate with future upgrades.

Switches (Cisco 3650 Multilayer Switches): The addition of these multilayer switches improves data transmission between different network layers (access, distribution, and core). They enhance both redundancy and speed while also simplifying network management by offering Layer 3 switching functionalities.

Wireless LAN Controller (Cisco 8500 Series): By centralizing the control of access points, this controller ensures seamless wireless connectivity across departments. It also enhances security and simplifies troubleshooting, making wireless network management more efficient.

Backup Server (Cisco UCS C240 M6 Rack Server): Adding a dedicated backup server improves data redundancy and disaster recovery capabilities. It ensures that critical data is regularly backed up and can be restored quickly, reducing downtime in case of failure.

Access Points (Cisco Aironet 3800 Series LWAP): The new access points improve wireless coverage across the network, providing users with better and more reliable wireless connections. This ensures that even high-density areas have consistent wireless performance.

The overall impact of these devices is an improved, scalable, and secure network that can handle increased traffic, enhance performance, and ensure better connectivity and data protection across all locations

Devices to be purchased:

1) Routers 4331 x2
2) Multilayer Switch 3650 x2
3) Cisco 8500 Series Wireless Lan Controller x1
4) Cisco UCS C240 M6 Rack server (Backup server) x1
5) Access/Layer-2 switches (48 ports) x1
6) Cisco Aironet 3800 Series (LWAP) x6
7) Cisco 7960 IP phone x80
8) PCs x135

No Devices will be redeployed.